

MOHAMMED LATIF SIDDIQ

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[GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ResearchGate](#)

EXECUTIVE SUMMARY

- Software engineering and applied large language models researcher with five years of hands-on experience.
- Industry-level software engineering experience developing enterprise and machine learning-based software.
- Published 20+ papers in top-tier venues and research applied at Meta Platform.
- **Research Interest:** Applied Language Models, Software Engineering, Software Security, and Testing.

EDUCATION

University of Notre Dame, USA

January, 2022 - May, 2026

Ph.D. in Computer Science and Engineering

Advisor: Dr. Joanna C.S. Santos

Thesis: Trustworthy Integration of Large Language Models in Software

Engineering: Risks, Evaluation, and Mitigation

University of Notre Dame, USA

January, 2022 - January, 2025

M.S. in Computer Science and Engineering (Non-thesis)

Advisor: Dr. Joanna C.S. Santos

Bangladesh University of Engineering and Technology, Bangladesh

February, 2016 - February, 2021

Bachelor of Science in Computer Science and Engineering

Advisor: Dr. Anindya Iqbal

Thesis: SQLIFIX: Learning Based Approach to Fix SQL Injection Vulnerabilities
in Source Code

PROFESSIONAL EXPERIENCES

Meta Platforms, CA, USA | *Research Scientist*

June, 2026 - Present

- Working with the WhatsApp core messaging team.
- **Tech stack:** Java, Kotlin, Swift, Objective-C, CodeQL, and JavaScript.

University of Notre Dame, IN, USA | *Graduate Assistant*

January, 2022 - May, 2026

- Authored 20+ peer-reviewed publications, focusing on applied LLMs software engineering and security.
- Mentored 25+ students, contributing to 10+ research outputs and successful graduate school admissions.

Meta Platforms, CA, USA | *Software Engineer Intern (Ph.D.)*

May, 2025 - August, 2025

- Worked with the WhatsApp iOS development team for groups and communities consumer messaging.
- Launched three projects in beta testing and lift 0.4% group add members worldwide.
- **Tech stack:** Swift, Objective-C, CodeQL, and JavaScript.

REVE Systems, Dhaka, Bangladesh | *Junior Software Engineer*

March, 2021 - November, 2021

- Built enterprise-grade applications for the Bangladesh Parliament Secretariat using Java-based technologies.
- **Tech stack:** Java servlet, JSP and MySQL.

PUBLICATIONS

Peer-reviewed full journal papers are prefixed with J, peer-reviewed full conference papers with C, peer-reviewed special track papers with S, peer-reviewed workshop papers with W, and preprints with P.

- ⟨C8⟩ **Mohammed Latif Siddiq**, Md. Nafiu Rahman, and Joanna C. S. Santos. “**An Empirical Study of Security Calibration in Large Language Models for Code**”. *42nd IEEE International Conference on Software Maintenance and Evolution (ICSME 2026)*.
- ⟨J6⟩ **Mohammed Latif Siddiq**, Tanzim Hossain Romel, Natalie Sekerak, Beatrice Casey, Joanna Santos. “**An Empirical Study on Remote Code Execution in Machine Learning Model Hosting Ecosystems**”. *ACM Transactions on Software Engineering and Methodology (Under-review)*.
- ⟨J5⟩ **Mohammed Latif Siddiq**, Xinye Zhao, Vinicius Carvalho Lopes, Beatrice Casey, and Joanna C. S. Santos. “**Security in the Age of AI Teammates: An Empirical Study of Agentic Pull Requests on GitHub**”. *Information and Software Technology (Under-review)*.
- ⟨J4⟩ **Mohammed Latif Siddiq**, Natalie Sekerak, Antonio Karam, Maria Leal, Arvin Islam-Gomes, and Joanna C. S. Santos. “**Assessing the Software Security Comprehension of Large Language Models**”. *Empirical Software Engineering (Under-review)*.
- ⟨J3⟩ **Mohammed Latif Siddiq**, Arvin Islam-Gomes, Natalie Sekerak, and Joanna C. S. Santos. “**Large Language Models for Software Engineering: A Reproducibility Crisis**”. *Empirical Software Engineering (Under-review)*.
- ⟨J2⟩ **Mohammed Latif Siddiq**, Noshin Ulfat, Nishat Raihan, Joanna C. S. Santos, and Marcos Zampieri. “**Multi-SALLM: A Multilingual Security Assessment of Generated Code**”. *Automated Software Engineering Journal (Under-review)*.
- ⟨J1⟩ Nishat Raihan, Dhiman Goswami, Sadiya Sayara Chowdhury Puspo, **Mohammed Latif Siddiq**, Christian Newman, Tharindu Ranasinghe, Joanna C.S. Santos, and Marcos Zampieri “**On the Performance of Large Language Models on Introductory Programming Assignments**”. *Journal of Intelligent Information Systems*.
- ⟨C7⟩ Nishat Raihan, **Mohammed Latif Siddiq**, Joanna C. S. Santos, and Marcos Zampieri. “**Large Language Models in Computer Science Education: A Systematic Literature Review**”. *56th ACM Technical Symposium on Computer Science Education (SIGCSE TS 2025)*.
- ⟨S3⟩ **Mohammed Latif Siddiq**. “**Advancing Secure and Standard Source Code Generation Techniques**”. *47th International Conference on Software Engineering, Doctoral Symposium (ICSE DS 2025)*.
- ⟨W5⟩ Mushfiquir Rahman and **Mohammed Latif Siddiq**. “**Code Comment Classification with Data Augmentation and Transformer-Based Models**”. *4th International Workshop on NL-based Software Engineering (NLBSE 2025)*.
- ⟨C6⟩ **Mohammed Latif Siddiq***, Jiahao Zhang*, and Joanna C. S. Santos (*Equal contribution). “**Understanding Regular Expression Denial of Service (ReDoS): Insights from LLM-Generated Regexes and Developer Forums**”. *32nd IEEE/ACM International Conference on Program Comprehension (ICPC 2024)*.
- ⟨C5⟩ **Mohammed Latif Siddiq**, Joanna C. S. Santos, Ridwanul Hasan Tanvir, Noshin Ulfat, Fahmid Al Rifat, and Vinicius Carvalho Lopes. “**Using Large Language Models to Generate JUnit Tests: An Empirical Study**”. *28th International Conference on Evaluation and Assessment in Software Engineering (EASE 2024)*.
- ⟨C4⟩ **Mohammed Latif Siddiq**, Simantika Dristi, Joy Saha, and Joanna C. S. Santos. “**The Fault in Our Stars: Quality Assessment of Prompts Used in Code Generation**”. *24th IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM 2024)*.

- ⟨C3⟩ **Mohammed Latif Siddiq**, Beatrice Casey, and Joanna C. S. Santos. “**FRANC: A Lightweight Framework for High-Quality Code Generation**”. 24th *IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM 2024)*.
- ⟨S2⟩ **Mohammed Latif Siddiq**, Lindsay Roney, Jiahao Zhang, and Joanna C. S. Santos. “**Quality Assessment of ChatGPT Generated Code and their Use by Developers**”. 21st *International Conference on Mining Software Repositories, Mining Challenge Track (MSR-Mining Challenge 2024)*.
- ⟨S1⟩ **Mohammed Latif Siddiq**, Jiahao Zhang, Lindsay Roney, and Joanna C. S. Santos. “**Re(gEx|DoS)Eval: Evaluating Generated Regular Expressions and their Proneness to DoS Attacks**”. 46th *International Conference on Software Engineering, New Ideas and Emerging Results Track (ICSE-NIER 2024)*.
- ⟨W4⟩ **Mohammed Latif Siddiq**, Joanna C. S. Santos, Sajith Devareddy, and Anna Muller. “**SALLM: Security Assessment of Generated Code**”. 6th *International Workshop on Automated and Verifiable Software System Development (ASYDE 2024)*.
- ⟨P2⟩ Rishov Paul, Md. Mohib Hossain, **Mohammed Latif Siddiq**, Masum Hasan, Anindya Iqbal, Joanna C. S. Santos. “**Enhancing Automated Program Repair through Fine-tuning and Prompt Engineering**”. *arXiv preprint arXiv:2304.07840*. 2023.
- ⟨W3⟩ **Mohammed Latif Siddiq**, Abdus Samee, Sk Ruhul Azgor, Md. Asif Haider, Shehabul Islam Sawraz, and Joanna C. S. Santos. “**Zero-shot Prompting for Code Complexity Prediction Using GitHub Copilot**”. 2nd *International Workshop on Natural Language-based Software Engineering (NLBSE 2023)*.
- ⟨C2⟩ **Mohammed Latif Siddiq**, Shafayat Hossain Majumder, Maisha Rahman Mim, Sourov Jajodia, and Joanna C. S. Santos. “**An Empirical Study of Code Smells in Transformer-based Code Generation Techniques**”. 22nd *IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM 2022)*.
- ⟨W2⟩ **Mohammed Latif Siddiq**, and Joanna C. S. Santos. “**SecurityEval Dataset: Mining Vulnerability Examples to Evaluate Machine Learning-Based Code Generation Techniques**”. 1st *International Workshop on Mining Software Repositories Applications for Privacy and Security (MSR4P&S 2022)*.
- ⟨W1⟩ **Mohammed Latif Siddiq**, and Joanna C. S. Santos. “**BERT-Based GitHub Issue Report Classification**”. 1st *International Workshop on Natural Language-based Software Engineering (NLBSE 2022)*.
- ⟨P1⟩ Waqar Hassan Khan, Md Al Imran, Ahmed Nafis Fuad, **Mohammed Latif Siddiq**, and ABM Islam. “**Shashthosheba: Dissecting Perception of Bangladeshi People towards Telemedicine Apps through the Lens of Features of the Apps**”. *arXiv preprint arXiv:2205.02793*. 2022.
- ⟨C1⟩ **Mohammed Latif Siddiq***, Md. Rezwanur Rahman Jahin*, Mohammad Rafid Ul Islam, Rifat Shahriyar, and Anindya Iqbal (*Equal contribution). “**SQLIFIX: Learning Based Approach to Fix SQL Injection Vulnerabilities in Source Code**”. 28th *IEEE International Conference on Software Analysis, Evolution and Re-engineering (SANER 2021)*.

SERVICES

PC Member, Emerging Results, Vision, and Reflection Papers Track, The International Symposium on Empirical Software Engineering and Measurement (ESEM-EVR), 2026

PC member, Deep Learning for Code Workshop (DL4C), 2026

PC Member, Ideas, Visions and Reflections Track, Foundations of Software Engineering (FSE-IVR), 2026

Committee Member, Artifact Evaluation Committee, Conference on Software Engineering (ICSE), 2026

Reviewer, Conference on Human-Computer Interaction (CHI), 2026

Journal Reviewer, Elsevier Information and Software Technology (IST), 2025 - 26

Journal Reviewer, Elsevier Journal of Systems and Software (JSS), 2025 - 26

Journal Reviewer, Springer Nature Empirical Software Engineering (EMSE), 2026
PC member, Workshop on Natural Language-based Software Engineering (NLBSE), 2026
Journal Reviewer, IEEE Transactions on Software Engineering (TSE), 2025
Journal Reviewer, Elsevier Cyber Security and Applications (CSA), 2025
PC member, Workshop on the use of Large Language Models for Cybersecurity (LLM4Sec), 2025
PC member, Deep Learning for Code Workshop (DL4C), 2025
Journal Reviewer, Elsevier Science of Computer Programming, 2024 - 25
Committee Member, Artifact Evaluation Committee, Conference on Software Engineering (ICSE), 2025
Junior PC member, Mining Software Repositories Conference (MSR), 2025
Research Mentor, CSE Summer Enrichment Program, Summer 2024
PC member, Automated Software Engineering Conference, Industry Showcase (ASE), 2024
Junior PC member, Mining Software Repositories Conference (MSR), 2024
Committee Member, Artifact Evaluation, Symposium on Software Testing and Analysis (ISSTA), 2024
Vice President, Bangladeshi Students Association of Notre Dame (BDSA-ND), 2024 - 25
Treasurer, Bangladeshi Students Association of Notre Dame (BDSA-ND), 2023 - 24
Student volunteer, Conference on Software Engineering (ICSE), 2022

MENTORSHIP EXPERIENCE

All mentees were supervised in conjunction with Dr. Joanna C.S. Santos (University of Notre Dame).

* Mentees from University of Notre Dame, IN, USA.

@ Mentees from Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.

Spring 2026

Natalie Sekerak*, Sajith Devareddy*, Arvin Islam-Gomes*, and Md. Nafiu Rahman@

Fall 2025

Tanzim Hossain Romel@, Arvin Islam-Gomes* and Natalie Sekerak*

Spring 2025

Antonio Karam*, Sajith Devareddy*, Maria C. Leal*, Neol Gutierrez*, Arvin Islam-Gomes* and Abby Kleist*

Fall 2024

Antonio Karam*, Sajith Devareddy*, Maria C. Leal*, Neol Gutierrez*, and Noshin Ulfat@

Summer 2024

AJ Jones*, and Vérité Mugabo*

Spring 2024

Lindsay Roney*, Sajith Devareddy*, Miguel Jose Maninang*, and Neol Gutierrez*

Fall 2023

Jiahao Zhang*, Anna Muller*, and Lindsay Roney*

Summer 2023

Simantika Dristi@, and Joy Saha@

Spring 2023

Abdus Samee@, Sk Ruhul Azgor@, Md. Asif Haider@, and Shehabul Islam Sawraz@

Fall 2022

Ridwanul Hasan Tanvir@, Noshin Ulfat@, and Fahmid Al Rifat@

Summer 2022

Shafayat Hossain Majumder@, Maisha Rahman Mim@, and Sourov Jajodia@

Spring 2022

Mark Cheng*

GRANTS & SCHOLARSHIPS

Research Grant, Google, 2026

1,000\$ Google Cloud credit to conduct software engineering research.

Travel Grants, NSF ICSE 2025 Travel Awards, 2025

1,513\$ grants to participate in ICSE 2025 in-person and Doctoral Symposium in Ottawa, Canada.

Travel Grants, Zahm Professional Development Fund, 2025

1,250\$ grants to participate in ICSE 2025 in-person in Ottawa, Canada.

Travel Grants, ACM SIGSOFT Travel Grants, 2025

550\$ grants to participate in ICSE 2025 in-person in Ottawa, Canada.

Travel Grants, ACM SIGSOFT ASE 2024 Travel Awards, 2024

600\$ grants to participate in ASE 2024 in-person in Sacramento, California, United States.

Travel Grants, NSF ICSE 2024 Travel Awards, 2024

1,938\$ grants to participate in ICSE 2024 in-person and Student Mentoring Workshop in Lisbon, Portugal with a 30% acceptance rate.

Travel Grants, ACM SIGSOFT Travel Grants, 2023

1,200\$ grants to participate in ICSE 2023 in-person and Student Mentoring Workshop in Melbourne, Australia with a 50% acceptance rate.

Travel Grants, Bangladesh Sweden Trust Fund, 2022 - 2023

100\$ grants for one-way plane fare to attend USA school with financial funding.

Travel Grants, NSF ICSE 2022 Travel Awards, 2022

1,075\$ grants to participate in ICSE 2022 in-person and Student Mentoring Workshop along with complimentary registration for the conference.

Research Assistantship, University of Notre Dame, 2022 - 2027

Tuition scholarship, including health insurance and bimonthly stipends.

Technical Scholarship, Bangladesh, 2016 - 2020

Complimentary government scholarship for regular engineering students.

Government Scholarship, Bangladesh, 2016 - 2020

Awarded for outstanding Performance in Higher School Certificate Examination.

Government Scholarship, Bangladesh, 2011 - 2012

Awarded for outstanding Performance in Junior School Certificate Examination.

ACHIEVEMENTS

U.S. EB-1A Extraordinary Ability Immigrant Petition Approved, 2025

Nominated for IBM Ph.D. Fellowship, 2024, 1 of 4 ND-CSE Ph.D. Students

Nominated for Google Ph.D. Fellowship, 2024, 1 of 4 ND-CSE Ph.D. Students

AI for Bangla, Bangladesh, 2021, Top 30

Google Hashcode, Global, 2019, 5th Among Bangladeshi Teams

IEEEEXtreme 12.0, Global, 2018, 4th Among Bangladeshi Teams

AUB Programming Contest, Bangladesh, 2018, 11th

Samsung Coding Contest, Bangladesh, 2018–2019, Finalist

Tech for Peace Hackathon, Bangladesh, 2017, Winner

TECHNICAL SKILLS

Languages: Java, C, C++, Python, Swift, Objective-C, Matlab, Dart, Shell

Database: Oracle, MySQL, SQLite, MongoDB, Firebase

Machine Learning Tools: Gradio, Langchain, Databricks, PyTorch

Version Control: Git(GitHub, Bitbucket, Gitlab), Sapling (Meta), TFS(Azure DevOps)

Frameworks: Vue.js, JSP, Flutter, JavaFX, Node.js, Django, Java servlet

Web Technology: HTML, CSS, Rest API, JSON, XML, GraphQL

Cloud: EC2, S3 Bucket, Azure

Operating Systems: Windows, Ubuntu 20.04, Windows Server 2016, Mac OS, CentOS 7/RedHat

Technical Writing: L^AT_EX, Beamer, Overleaf

Miscellaneous: Google Apps Script, Software Defined Networking, OpenGL, Weka, Mininet, NS2

REFERENCE

Last update: June 4, 2026

Dr. Joanna Cecilia da Silva Santos

Assistant Professor

Department of Computer Science and Engineering

University of Notre Dame, IN, USA

Email: joannacss@nd.edu

Relation: Ph.D. Advisor

Dr. Marcos Zampieri

Assistant Professor

School of Computing

George Mason University, Fairfax, VA, USA

Email: mzampier@gmu.edu

Relation: Research Collaborator